Example 2.2

To study about the introduction of NEET exam, the opinions are collected from 3 schools. The strength of the schools are 2000, 2500 and 4000. It is fixed that the sample size is 170. Calculate the sample size for each school?

Solution

Here $N = 2000 + 2500 + 4000 = 8500$ and $n = 170$ then $n_1 = n_2 = n_{13} = ?$

$N_1 = 2000, \quad N_2 = 2500, \quad N_3 = 4000$

$$n_1 = (n/N) \times N_1 = (170 / 8500) \times 2000 = 40$$

$$n_2 = (n/N) \times N_2 = (170 / 8500) \times 2500 = 50$$

$$n_3 = (n/N) \times N_3 = (170 / 8500) \times 4000 = 80$$

Therefore 40 students from school 1, 50 students from school 2 and 80 students from school 3, are to be selected using SRS to obtain the required stratified random sample

The main objective of stratification is to give a better cross-section of the population for a higher degree of relative precision. The criteria used for stratification are States, age and sex, academic ability, marital status etc,. In many practical situations when it is difficult to stratify with respect to the characteristic under study, administrative convenience may be considered as the basis for stratification

Merits

- It provides a chance to study of all the sub-populations separately.
- An optimum size of the sample can be determined with a given cost, precision and reliability.
- It is a more precise sample.
- Representation of sub groups in the population
- Biases reduced and greater precise.

Limitations

- There is a possibility of faulty stratification and hence the accuracy may be lost.
- Proportionate stratification requires accurate information on the proportion of population in each stratum.

2.7.3 Systematic Sampling

In systematic sampling, the population units are numbered from 1 to N in ascending order. A sampling interval, denoted by k, is determined as $k = \frac{N}{n}$, where n denotes the

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